SYSTEMATIC EQUITY • MARKET-NEUTRAL STOCK SELECTION • OPEN-SOURCE PROOF OF CONCEPT

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First-Mover Flow (FMF-N)

Zero-beta alpha for concentrated equity portfolios, from *who* revises first and *when*

THE BUSINESS PROBLEM

Tech, thematic and high-conviction funds, and QIS baskets on narrow universes, all run **concentrated portfolios**. Three things go wrong there:

- **Too few bets:** With 10–30 names, breadth is tiny and real stock-picking skill is statistically hard to see.
- **One dominant factor:** Most signals turn out to be beta or momentum in disguise.
- **Overlays manufacture Calmar:** A regime filter that adds net exposure in a bull market looks like skill. Our own first-generation strategy reached a Calmar of 1.77, but its timing leg underperformed a *static* beta of the same size.

What's needed: stock-selection alpha that is *independent of the market*, that raises the Calmar ratio on its own, and that blends cleanly into the funds and indices desks already run.

THE IDEA

Sell-side analysts do not revise at the same time. **First movers**, who revise on an earnings day or after ≥10 silent sessions, carry information the market absorbs over weeks. **Herders**, who copy an earnings cluster 3–15 sessions later, mark exhaustion. The novel **leader-minus-herd** signal (LMH) subtracts one from the other:

- leaders: IC **+0.066**, herders: IC −0.084
- the only one of 18 candidates with power beyond four price factors (Fama–MacBeth t = 2.46)
- **replicates in a never-seen period**: IC **0.096**, p = 0.042, 0.105 on the 20 names never used in development

LMH is combined with three revision-breadth signals in an equal-weight rank composite: no fitted weights, nothing to overfit.

THE RESEARCH PATH: FROM REGIME TIMING TO SELECTION ALPHA

Gen 1: Regime-Timed Breadth (RTB)
Classic revision breadth, GJR-GARCH sizing and an HMM/trend regime tilt on net exposure. Point in time and walk-forward. **Calmar 1.77** (development), 1.65 (holdout).

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Diagnosis: beta camouflage
The selection book alone: Sharpe 0.63, Calmar 0.36, placebo p 0.08–0.14. A *constant* 0.2 beta beat the timing leg (Calmar 1.82). Little stock-picking alpha was left once beta was removed.

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Gen 2: First-Mover Flow (FMF-N, FMF-B20)
New signal (leader minus herd), 42-session tranches, zero beta. Selection Sharpe 0.63→0.98, Calmar 0.36→0.79, turnover 10.8x→2.1x, placebo p = 0.022.

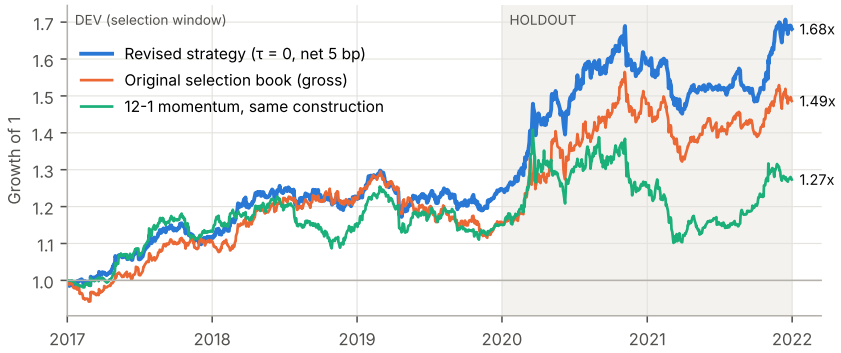
IN SAMPLE 2017–2021 (10 US large caps, development + holdout)

1.13	0.78	−0.02	7.4%
Sharpe, net	**Calmar**	**Beta**	**4-factor alpha**
holdout 1.33	Gen 1 book 0.53	dollar-neutral	t = 2.98

OUT OF SAMPLE 2024–2026 (30 stocks, never-seen period, frozen, pre-registered)

1.05	1.32	11.4%	#2 / 14
Sharpe, net	**Calmar**	**CAPM alpha**	**vs institutional**
both halves > 0	max DD −8.0%	t = 2.04	on both metrics

Dollar-neutral stock selection, 2017–2021



OUTPERFORMANCE AGAINST THE INSTITUTIONAL TOOLKIT (SAME CONSTRUCTION, NET OF 5 BP)

	In sample, 2017–2021				Never-seen period, 2024–26	
	Sharpe	Calmar	Max DD	Corr. FMF	Sharpe	Calmar
First-Mover Flow, FMF-N	**1.13**	**0.78**	−14.1%	1.00	**1.05**	**1.32**
12-1 momentum (Jegadeesh–Titman)	0.50	0.23	−21.8%	0.71	0.60	0.62
Residual momentum (Blitz et al.)	0.17	0.05	−24.1%	0.64	0.56	0.54
Low volatility	0.56	0.37	−13.6%	0.04	−0.39	−0.17
Betting-against-beta	−0.15	−0.07	−32.2%	0.10	−0.94	−0.32
Consensus target-price upside	0.38	0.24	−12.1%	−0.38	0.11	0.05
Classic revision breadth	0.92	0.58	−15.5%	0.95	0.98	1.16
Multi-factor: momentum + low vol + reversal	0.81	0.42	−14.3%	0.44	−0.00	−0.01
First-Mover Flow + 0.2 beta budget, FMF-B20	1.62	**1.66**	−10.8%	–	1.49†	**2.43**†
Regime-Timed Breadth, RTB (Gen 1)	1.48	1.39	−13.6%	–	–	–
Equal-weight long-only market (beta 1)	1.28	0.87	−38.1%	−0.02	1.14	0.94

FMF-N ranks first of 14 cross-sectional books in sample and second out of sample (behind target-price momentum, Sharpe 1.14). A 50/50 blend with FMF-N raises the Sharpe of 16/18 institutional books in sample and 16/16 out of sample. All choices made on 2017–19 only. †FMF-B20 out of sample is exploratory: its 0.2 budget was fixed in development but not listed in the pre-registration.

HOW IT WORKS: SIGNAL, CONSTRUCTION AND RISK LOGIC

Raw data daily closes, analyst target prices → **Point in time** usable the session after publication, expiry 252 sessions, audited cleaning → **4 revision-flow signals** breadth, event breadth, breadth ex-momentum, **leader minus herd** → **Composite** equal-weight ranks, no fitted weights → **Book** long top / short bottom 40%, inverse GJR-GARCH vol in each leg → **42 daily tranches** 1/42 of the book re-targeted each day, net exposure 0 (or a fixed budget)

RISK-MANAGEMENT LOGIC

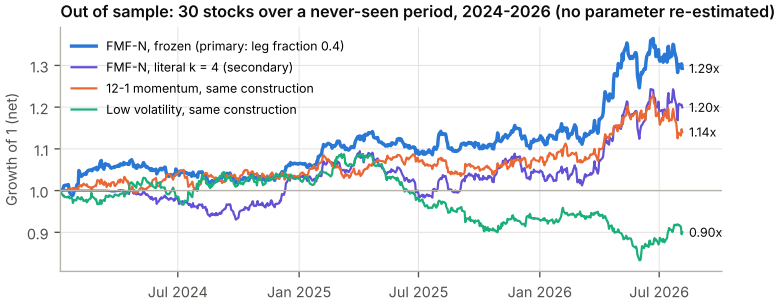
- **Beta is a decision, not a by-product:** The book is dollar-neutral every day. Any market exposure is a constant, disclosed beta budget, never a timed overlay. FMF-B20 (net 0.2) reaches a Calmar of 1.66 over 2017–21, against 1.39 for the regime-timed predecessor, with a max drawdown of −10.8%.
- **Volatility-aware sizing:** Inverse GJR-GARCH-t weights shrink the names whose drawdowns would dominate.
- **Horizon-matched trading:** The information builds over weeks (IC 0.026 at 1 day, 0.109 at 21). Overlapping tranches remove rebalancing-date luck: a single weekly rebalance date gives a Sharpe anywhere from 0.34 to 0.71 for the same signal. Turnover is 2.1–2.7x per year, against 10.8x before.
- **Costs and capacity:** All results are net of 5 bp per unit traded (Sharpe 1.11 at 10 bp). The universe is liquid large caps.

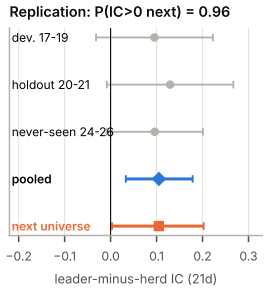
CRISIS BEHAVIOUR (CUMULATIVE RETURN, %)

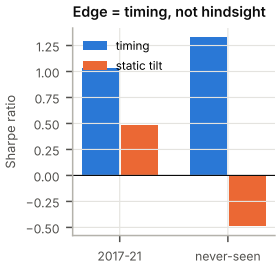
	COVID crash	Vaccine rally	Growth sell-off
	Feb–Mar 20	Nov 20	Feb–Mar 21
FMF-N	**+12.0**	**−5.8**	**−7.1**
12-1 momentum	+12.2	−8.2	−10.5
Betting-against-beta	−15.2	−1.0	−1.7
Trend following (CTA)	−18.9	−3.5	−12.1
Equal-weight market	−35.5	+6.9	−9.4

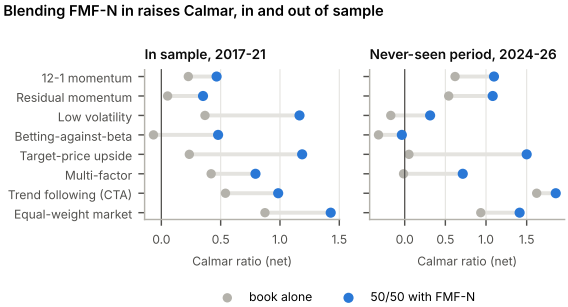
ATTACKED FROM NINE ANGLES, AS AN INVESTMENT COMMITTEE WOULD

Objection	Test	Result
"Hindsight universe: it is just long the winners."	Causal split into static tilt and timing books (Lo 2008)	Timing earns Sharpe 1.04 (t 2.41), never-seen 1.33 (t 2.21). Static tilt: −0.49.
"Close-of-day fills and free shorts are fantasy."	0–2 day delay × 0–200 bp borrow × 0–30 bp costs	Sharpe 1.06 at 1 day, 100 bp, 10 bp, breakeven cost 234 bp per unit traded.
"Dollar-neutral is not beta-neutral."	Separate up- and down-market betas, weekly	Down-market beta −0.07 (t −1.24), never-seen −0.01.
"Overlapping tranches smooth the P&L."	Lo (2002) autocorrelation-adjusted Sharpe	1.06 in both samples (lag-1 autocorrelation 0.04).
"You chose the parameters that work."	Multiverse of ∼300 specifications (incl. quarterly expiry)	99% positive (97% out of sample), selection-free ensemble Sharpe 0.92.
"The new signal adds nothing to breadth."	Same book without leader-minus-herd, block bootstrap	+0.28 Sharpe (p 0.028), out of sample +0.30, pooled $P(>0)$ 0.94.
"Beating the toolkit is cherry-picking."	Romano–Wolf vs 16 books, reverse SPA test	No comparable beats FMF-N (SPA p 0.86 / 0.86).
"Ten stocks prove nothing."	Bayesian meta-analysis of 3 non-overlapping samples	P(signal works in the **next** universe) = 0.96, Stouffer p = 0.0022.
"One history, one drawdown."	Diffusion-FHS generator (classifier score 0.014, 0 = indistinguishable)	5-year median Calmar 0.70, 5% max DD −34%, momentum crashes are the binding stress.









WHO IT IS FOR

- **PMs** of concentrated books: a zero-beta selection sleeve, with today's book, trade list, risk report and signal monitor.
- **QIS structurers**: a rules-based index (excess return, vol target, fees, generated rulebook). Blended 50/50 it raises the Calmar of every price-factor index tested.
- **Quant researchers**: a signal registry, a multiverse, Bayesian and snooping-robust tests and generative stress tests, all runnable on your own data.

RIGOR AND HONEST LIMITS

Point in time and truncation-tested, out-of-sample rules hashed before any result, more than fifty failed ideas documented. Deflated over every configuration tried, the development Sharpe alone is not conclusive (DSR 0.18–0.38), and no single short window is significant once the whole family of specifications is accounted for. The evidence is replication across three non-overlapping periods, ten of the 30 out-of-sample names are development names, and the 20 others alone give a Sharpe of 0.80. **Code:** `fmflow`, 100+ tests, 11 notebooks, Docker, it reproduces the research notebook exactly.

Khaled Baha • Khaled.Cheikh.Baha@gmail.com • github.com/bahakh19/first-mover-flow • Paper: "Who Revises First? Analyst Revision Flow and Zero-Beta Stock Selection in Concentrated Equity Portfolios". Past performance, in or out of sample, does not guarantee future results.